

MANISH RAINA

AI/ML Engineer

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PROFILE

AI/ML Engineer with hands-on, end-to-end project experience across model development, backend engineering, and full-stack delivery. Proven track record shipping multi-agent orchestration systems, production ML pipelines, and full-stack apps with measurable accuracy, latency, and reliability gains using Python, Scikit-learn, TensorFlow, FastAPI, React, Docker, and RAG/MCP-based agent systems.

CORE COMPETENCIES

- **Languages:** Python, C, C++, JavaScript, SQL, TypeScript
- **AI Agents & Orchestration:** LangChain, LangGraph, Model Context Protocol (MCP), Multi-Agent Systems, Agentic Tool/Skill Design, Function Calling, Prompt Engineering, RAG Pipelines
- **ML & Model Training:** Scikit-learn, TensorFlow, LightGBM, NLP, Model Training & Fine-Tuning, Embedding-based Semantic Search, Model Evaluation (AUC, Cross-Validation, Calibration, IPCW Weighting)
- **Web & Backend / APIs:** FastAPI, React.js, REST APIs, API Design & Integration, WebSockets, JWT Authentication, RBAC, Middleware Validation
- **Data & Infra:** PostgreSQL, MongoDB, Redis, SQLite, Qdrant Vector DB, Docker, Docker-compose, Git
- **Other:** Web Scraping, Power BI, Speech Recognition & Synthesis, OpenCV, dlib, Playwright

EXPERIENCE

Machine Learning & Web Development Intern — EISystems Technologies Aug 2024 – Sept 2024

- Worked on data preprocessing and ML model development for client projects, alongside front-end feature development using React.js.

Machine Learning Research Intern — NIT Delhi Aug 2025 – Sept 2025

- Researched and implemented supervised machine learning algorithms using Python, Scikit-learn, Pandas, and NumPy; conducted comparative performance analysis and prepared technical documentation.

PROJECTS

Nexus AI OS v2.0 — Autonomous Multi-Agent Workspace & AI Orchestration Platform [GitHub](#)

- Engineered a multi-agent orchestration kernel routing tasks across **12 specialized agents** via a JSON-based DAG, supporting **12 task categories and 30+ sub-actions**, with a custom **MCP client** integrating **6 MCP servers** and **10 built-in tools**.
- Achieved a **100% pass rate** on integration tests (4/4) and DAG workflows (6/6), with **96% task-step success** (24/25); full 4-step workflows completed in **3.8–4.2 seconds**.

AI Career Skill Intelligence Platform (Full-Stack + ML + Scraper) [GitHub](#)

- Architected a full-stack skill-gap platform (**FastAPI, React, PostgreSQL**) with **39 REST endpoints**, parsing **55 resumes at 98.2% extraction accuracy** and computing readiness scores in **65.3ms** end-to-end.
- Processed **32,217 job listings** across **3,947 companies**, feature-engineering **94,766 skills** and training a **RandomForest** skill-demand regressor (**$R^2 = 0.9998$**); generated readiness scores for **50+ candidates** (avg 64.5%, top 92%).

Heart Disease Survival Prediction Framework [GitHub](#)

- Built a survival prediction model (**LightGBM + IPCW**) on **897 patient records** with 13 clinical features, handling **67.89% censored observations** via inverse probability of censoring weighting.
- Achieved **91.89% mean cross-validation AUC** ($\pm 1.37\%$ across 5 folds, 75 tuning fit-runs); outperformed a Cox Proportional Hazards baseline by **15.46%** at the 180-day horizon (ROC-AUC 0.8534 to 0.9026).

AI Voice Receptionist (Speech-to-Speech AI Assistant) [GitHub](#)

- Built a speech-to-speech voice assistant achieving **88% intent recognition accuracy** across 50 test queries, with **1.2–1.6s** speech-to-response latency and offline **pyttsx3** TTS generating responses in **80–120ms**.
- Integrated a real-time face-recognition layer (**DeepFace, VGG-Face, OpenCV**) achieving **91.7% accuracy** across 60 live trials at ~450–650ms per inference frame, plus automated Gmail IMAP email briefing.

EDUCATION

B.Tech, Computer Science Engineering (AI & ML) — PIET, Haryana Aug 2022 – Aug 2026

Senior Secondary (Class XII) — Kotwal National Institute Mar 2018 – Mar 2022

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning with Python — Internship Certificate, **EISystems Technologies** [Certificate](#)
- Machine Learning Research Internship — **NIT Delhi** (2025) [Certificate](#)
- Advanced Diploma in Software Technology (12 Months) — Supertech (India) Computer Education [Certificate](#)